The Price of Location Privacy: Age of Incorrect Information in Location-Aided UAV Communications

Author information to be added

***Abstract*—Location information increasingly supports wireless resource allocation through location-aided channel prediction and beam selection. Protecting this information can affect transmission reliability and the delivery of correct task updates. We consider a single-antenna unmanned aerial vehicle (UAV) communicating with a multi-antenna base station. The UAV releases a three-dimensional Laplace-perturbed location, and the base station applies maximum a posteriori (MAP) inference before selecting a receive beam. For a prescribed beam map and symmetric binary task source, we relate privacy-dependent outage to average age of incorrect information (AoII). Location-Aware Privacy–Information Selection (LAPIS) selects a fixed reporting budget using simultaneous confidence bounds and independent confirmation. In a conditional empirical model constructed from measured channel state information (CSI), MAP lookup yields an estimated 15.8% lower AoII than direct-report lookup at one tested budget. With 200 selection samples per scenario and budget, point-estimate selection violates the AoII constraint in 65.7% of 1,000 trials, whereas LAPIS has no observed violations and selects larger disclosure budgets. Separate geometric controls show that larger arrays can increase sensitivity to location uncertainty. These results quantify a privacy–reliability tradeoff under explicit model assumptions. Code and reproducibility materials: https://github.com/jwentong/LocationPrivacy.**

***Index Terms*—Location privacy, geo-indistinguishability, location-aided beamforming, age of incorrect information, UAV communications.**

I. Introduction

Spatial intelligence is creating demand for networks that understand where devices operate and how their surroundings affect communication [1]. This is particularly relevant to the low-altitude economy, where unmanned aerial vehicle (UAV) applications require reliable links for sensing, inspection, and logistics in three-dimensional environments [2]. Location consequently becomes more than a navigation coordinate: it links a terminal to environmental knowledge and can guide wireless resource allocation.

Existing research provides several enabling components. Location-based beamforming learns resource decisions from spatial information, including in non-line-of-sight environments [3]. Neural radio-frequency radiance fields (NeRF^2) model radio propagation through a continuous field conditioned on transmitter location [4]. Wireless radiation field reconstruction with Gaussian splatting (WRF-GS) reconstructs spatial spectra using three-dimensional Gaussian splatting [5], while WRF-GS+ improves field representation and prediction [6]. These advances support reusable environmental models and can reduce repeated channel acquisition. Their resource decisions nevertheless depend on the quality of the reported location.

Such uncertainty arises naturally from privacy protection. Accurate location reports expose a device's whereabouts and activities. Geo-indistinguishability protects reports through distance-based differential privacy [7], including three-dimensional mechanisms [8]. Related work protects against physical-layer localization using transmit beamforming [9], studies age of information (AoI) with privacy-aware mobile agents [10], and mitigates location leakage in federated radio-map construction [11]. We examine their operational consequence: how does protecting the location used for resource decisions affect subsequent task-information updates?

Consider a UAV whose location report is used by a multi-antenna base station (BS) to configure a receive beam. Stronger protection can increase inference error and beam mismatch. Localization error alone cannot quantify the resulting cost: radial and angular errors affect array gain differently, and the link margin determines whether a gain loss causes an outage. Missed packets also need not imply an incorrect task estimate. Unlike packet staleness, age of incorrect information (AoII) measures how long that estimate remains incorrect [12]. Studies of imperfect channel information [13] and low-altitude multiple-input multiple-output (MIMO) networks [14] motivate connecting resource errors to persistent information errors.

We study this connection in a single-UAV uplink with one transmit antenna and multiple BS receive antennas. A three-dimensional Laplace mechanism protects reports; maximum a posteriori (MAP) inference determines a location-aided receive beam, which is evaluated on the true channel. A separate Markovian task source connects link outcomes to AoII. We protect the report interface and assume the controller has no independent precise radio-frequency (RF) localization. The challenge is to connect privacy, inference, directional gain, outage, and source evolution, rather than treating location distortion as communication loss. The contributions are:

- A model connecting three-dimensional report privacy and Bayesian inference to receive-beam decisions and actual transmission outage.
- An analysis that specializes established binary-source AoII models to location-dependent outages, capturing the occurrence and persistence of incorrect task estimates.
- Location-Aware Privacy–Information Selection (LAPIS), a finite-grid procedure combining established concentration bounds with independent confirmation under privacy or AoII requirements.

Experiments using measured channel state information (CSI)

and geometric controls quantify the tradeoff. At one tested budget, MAP lookup yields an estimated 15.8% lower AoII. With 200 selection samples per scenario and budget, point-estimate selection violates the AoII constraint in 65.7% of 1,000 trials, versus no observed violations for LAPIS, which selects larger disclosure budgets (Section V).

II. System Model

A. Network and Update Protocol

In Fig. 1, a single-antenna UAV communicates with a BS at $\mathbf{b} \in \mathbb{R}^3$ using an $M_x \times M_y$ uniform planar array (UPA), $M = M_x M_y$. The UAV hovers at $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^3 \setminus \{\mathbf{b}\}$ per session; locations define separate scenarios. Power P, array, and packet requirements are fixed across budgets.

Each slot lasts T. At its start, the UAV generates state S_t and a noisy location report. A reliable control path delivers the report before beam configuration and state-packet transmission. The state remains constant within the slot; reception is assessed at its end. Failed packets are discarded. Fixed control overhead leaves data duration τ_d, identically across budgets.

Before beam selection, the controller observes reports, the public prior, and a frozen beam map, but no independent precise location or full instantaneous array CSI. Privacy is guaranteed only for the report interface, not for RF localization observations. The task source is independent of location and channel randomness and carries no true location.

B. Private Location Reporting and Inference

For a fixed budget $\epsilon > 0$, the UAV releases $\mathbf{Z}_t = \mathbf{x} + \mathbf{N}_t$, independently each slot. Following geo-indistinguishability [7], [8], we use the three-dimensional isotropic Laplace density

$$f_\epsilon(\mathbf{z} \mid \mathbf{x}) = \frac{\epsilon^3}{8\pi} \exp(-\epsilon \|\mathbf{z} - \mathbf{x}\|_2). \tag{1}$$

The triangle inequality gives

$$f_\epsilon(\mathbf{z} \mid \mathbf{x}) \le e^{\epsilon \|\mathbf{x} - \mathbf{x}'\|_2} f_\epsilon(\mathbf{z} \mid \mathbf{x}'). \tag{2}$$

Thus $\eta = \epsilon r_0$ is the dimensionless single-report budget within a protection distance r_0; smaller η provides stronger protection. Here ϵ has units m^{-1}. Reports are not clipped to the candidate region. The noise radius satisfies $\|\mathbf{N}_t\|_2 \sim \mathrm{Gamma}(3, 1/\epsilon)$ in shape–scale notation, with mean $3/\epsilon$.

Let $\mathcal{X} = \{\mathbf{x}_j\}_{j=1}^{N_\mathrm{x}}$, with $N_\mathrm{x} = |\mathcal{X}|$, be a public candidate set with fixed prior probabilities $\pi_j > 0$, $\sum_j \pi_j = 1$. A report $\mathbf{z}_t$ yields posterior

$$\rho_t(j) = \frac{\pi_j e^{-\epsilon \|\mathbf{z}_t - \mathbf{x}_j\|_2}}{\sum_{k=1}^{N_\mathrm{x}} \pi_k e^{-\epsilon \|\mathbf{z}_t - \mathbf{x}_k\|_2}}, \quad \widehat{\mathbf{x}}_t = \mathbf{x}_{\arg\max_j \rho_t(j)}. \tag{3}$$

Ties use a fixed rule. The prior and candidate set are public and fixed before reporting. Current-report MAP is a prescribed receiver, not an optimal attacker or AoII-optimal beam rule; temporal filtering is excluded. Post-processing preserves (2). For K retained reports, the likelihood ratio is bounded by $e^{K\epsilon\|\mathbf{x}-\mathbf{x}'\|_2}$, so the per-report budget does not guarantee session-level privacy.

C. Location-Aided Receive Beamforming

We adapt location-to-beam mapping [3] to uplink combining. For wavelength λ_c and array-relative element locations $\mathbf{r}_m$, the far-field narrowband steering vector for $\mathbf{v} \neq \mathbf{b}$ is

$$\mathbf{u}(\mathbf{v}) = \frac{\mathbf{v} - \mathbf{b}}{\|\mathbf{v} - \mathbf{b}\|_2}, \qquad [\mathbf{a}(\mathbf{v})]_m = e^{\mathrm{j}(2\pi/\lambda_c)\mathbf{r}_m^\mathsf{T}\mathbf{u}(\mathbf{v})}. \tag{4}$$

For a UPA, $\mathbf{r}_{m_x,m_y} = (m_x d_x, m_y d_y, 0)^\mathsf{T}$, where $0 \le m_x < M_x$ and $0 \le m_y < M_y$, and all coordinates share the array frame. Hence $\|\mathbf{a}(\mathbf{v})\|_2^2 = M$. A Rician baseline is

$$\mathbf{h}_t(\mathbf{x}) = \sqrt{\beta(\mathbf{x})} \left(\sqrt{\frac{\kappa}{\kappa+1}} \mathbf{a}(\mathbf{x}) + \sqrt{\frac{1}{\kappa+1}} \mathbf{g}_t \right), \tag{5}$$

where $\mathbf{g}_t \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$ is independent and identically distributed (i.i.d.) across slots and independent of report noise, κ is the linear Rician factor, and $\beta(\mathbf{x}) = \beta_0(\|\mathbf{x} - \mathbf{b}\|_2/d_0)^{-\nu}$ is the single-element large-scale power gain, with $\beta_0, d_0 > 0$ and $\nu > 0$. The line-of-sight (LOS) limit is obtained as $\kappa \to \infty$.

Given a frozen map output $\boldsymbol{\psi}(\mathbf{v}) \in \mathbb{C}^M \setminus \{\mathbf{0}\}$, the receive beam is

$$\mathbf{w}_t = \mathcal{W}(\widehat{\mathbf{x}}_t) = \frac{\boldsymbol{\psi}(\widehat{\mathbf{x}}_t)}{\|\boldsymbol{\psi}(\widehat{\mathbf{x}}_t)\|_2}, \qquad \|\mathbf{w}_t\|_2 = 1. \tag{6}$$

The geometric model uses $\boldsymbol{\psi}(\mathbf{v}) = \mathbf{a}(\mathbf{v})$; the empirical map uses a covariance eigenvector (Section V). Channel predictions are also compatible, but scalar power alone cannot specify (6). A fixed unit beam handles zero outputs. The prescribed digital beam map has no constant-modulus constraint and is not jointly optimized with the budget.

For unit-energy data symbol s_t and receiver noise $\mathbf{n}_t^{\mathrm{RF}} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_M)$, independent of other randomness, the received signal and combined signal-to-noise ratio (SNR) are

$$\begin{aligned} \mathbf{y}_t &= \sqrt{P}\mathbf{h}_t(\mathbf{x}) s_t + \mathbf{n}_t^{\mathrm{RF}}, \\ \Gamma_t &= \frac{P|\mathbf{w}_t^\mathsf{H}\mathbf{h}_t(\mathbf{x})|^2}{\sigma^2}. \end{aligned} \tag{7}$$

Unit-norm combining leaves noise power σ^2. In pure LOS with the geometry-based rule, the mismatch factor and resulting SNR are

$$\begin{aligned} B_t &= \frac{|\mathbf{a}(\widehat{\mathbf{x}}_t)^\mathsf{H}\mathbf{a}(\mathbf{x})|^2}{M^2} \in [0,1], \\ \Gamma_t &= \Gamma_{\mathrm{al}} B_t, \qquad \Gamma_{\mathrm{al}} = P\beta(\mathbf{x})M/\sigma^2. \end{aligned} \tag{8}$$

D. Outage and Task-Information Correctness

With fixed packet length L bits and bandwidth W, we use the ideal coding threshold $\gamma_{\mathrm{th}} = 2^{L/(W\tau_d)} - 1 > 0$. Define

$$Y_t = \mathbf{1}\{\Gamma_t \ge \gamma_{\mathrm{th}}\}, \qquad p_\epsilon(\mathbf{x}) = \Pr(Y_t = 1 \mid \mathbf{x}, \epsilon), \tag{9}$$

and $q_\epsilon(\mathbf{x}) = 1 - p_\epsilon(\mathbf{x})$. Outcomes are i.i.d. conditional on a session location, averaging over report noise and fading. In pure LOS, $p_\epsilon(\mathbf{x}) = \Pr\{B_t \ge \gamma_{\mathrm{th}}/\Gamma_{\mathrm{al}}\}$. At $\kappa = 0$, isotropic scattering makes an independent unit-norm projection direction-invariant, eliminating this directional privacy cost.

Following binary-source AoII models [12], [13], let $S_t \in \{0,1\}$ be a symmetric Markov chain with $\Pr(S_t = 1 - s \mid$

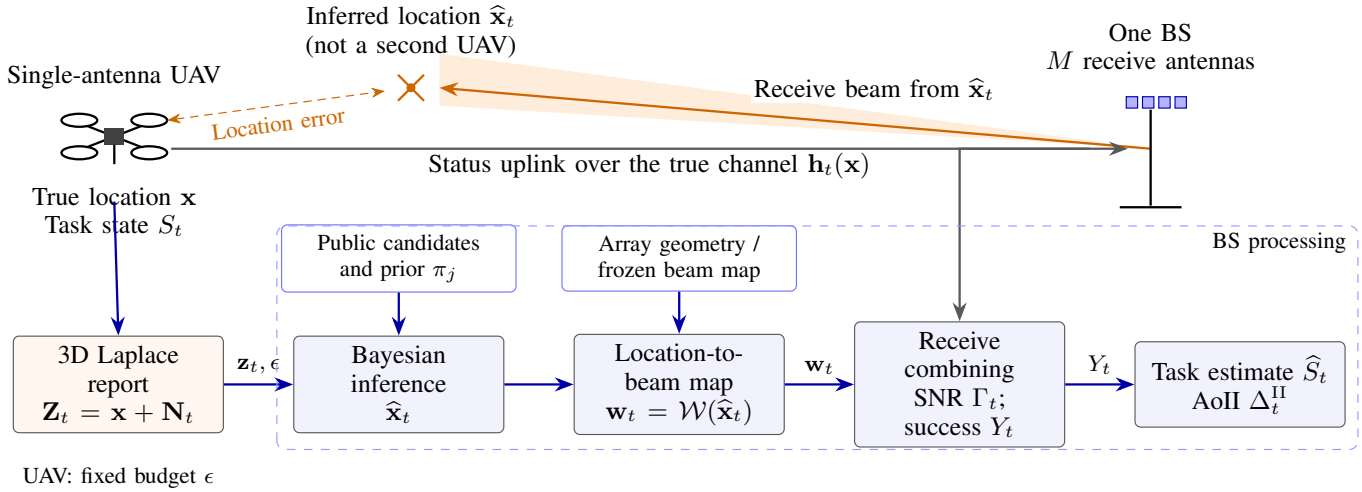


Fig. 1. Location-report privacy and its communication cost in a single-UAV uplink. The noisy report determines the inferred location and receive beam; the task packet propagates over the channel at the true location. The orange marker is an estimate, not another UAV. Location and task packets have distinct roles.

$S_{t-1} = s) = \alpha$ for $s \in \{0, 1\}$ and $0 < \alpha \leq 1/2$. Successful packets are decoded correctly; only threshold outage is modeled. At slot end,

$$\widehat{S}_t = \begin{cases} S_t, & Y_t = 1, \\ \widehat{S}_{t-1}, & Y_t = 0. \end{cases} \tag{10}$$

The linear AoII is $\Delta_t^{\mathrm{II}} = TA_t$, where

$$A_t = \begin{cases} 0, & S_t = \widehat{S}_t, \\ A_{t-1} + 1, & S_t \neq \widehat{S}_t. \end{cases} \tag{11}$$

Source reversal can restore correctness without reception, distinguishing this slot-end AoII from packet staleness.

III. Problem Formulation

A. From Outage to Average AoII

Specializing the binary-source framework of [12], [13] to our slot-end update convention, write $q = q_\epsilon(\mathbf{x})$. Since transmission outcomes are independent of the task source and past slots, (10)–(11) imply

$$\begin{aligned} \Pr(A_t = 1 \mid A_{t-1} = 0) &= \alpha q \triangleq \mu_0, \\ \Pr(A_t = a + 1 \mid A_{t-1} = a) &= (1 - \alpha) q \triangleq \mu_1, \quad a \geq 1. \end{aligned} \tag{12}$$

All other transitions return to zero. The stationary probabilities are $\xi_0 = (1-\mu_1)/(1-\mu_1+\mu_0)$ and $\xi_a = \xi_0 \mu_0 \mu_1^{a-1}$ for $a \geq 1$. Therefore,

$$\begin{aligned} \overline{\Delta}_\epsilon^{\mathrm{II}}(\mathbf{x}) &\triangleq \lim_{N\to\infty} \frac{1}{N} \mathbb{E}\left[\sum_{t=1}^{N} TA_t \mid \mathbf{x}\right] \\ &= T \frac{\alpha q_\epsilon(\mathbf{x})}{[1 - (1-\alpha) q_\epsilon(\mathbf{x})]\,[1 - (1-2\alpha) q_\epsilon(\mathbf{x})]}. \end{aligned} \tag{13}$$

For $0 < \alpha \leq 1/2$, (13) increases with q, and perfect transmission yields zero slot-end AoII. This does not establish monotonicity in ϵ.

B. Fixed Privacy-Budget Selection

Let $\mathcal{E} \subset (0, \infty)$ be a finite set of candidate budgets, fixed before testing. To select a common budget for a public distribution of hovering scenarios, specify weights $\omega_j \geq 0$, $\sum_j \omega_j = 1$, independently of the private test location, and define

$$J(\epsilon) = \sum_{j=1}^{N_{\mathrm{x}}} \omega_j \overline{\Delta}_\epsilon^{\mathrm{II}}(\mathbf{x}_j). \tag{14}$$

Weights describe separate sessions, not UAV motion, and differ in role from the MAP prior. AoII must be averaged after conditioning on location; applying (13) to average outage generally differs.

Given a maximum single-report budget $\eta_{\max}$ at distance r_0, the primary problem is

$$\begin{aligned} (\mathrm{P1}) \quad \min_{\epsilon \in \mathcal{E}} \quad & J(\epsilon) \\ \text{subject to} \quad & \epsilon r_0 \leq \eta_{\max}. \end{aligned} \tag{15}$$

Algorithm 1 selects ϵ offline for the entire session. The stationary objective does not imply a bounded lifetime privacy guarantee. If a finite session of K reports instead has total budget B_{priv} under the repeated-location model, replace the constraint by $K\epsilon r_0 \leq B_{\mathrm{priv}}$. This is static selection, not temporal budget allocation.

An alternative design perspective prioritizes the strongest report protection compatible with an AoII requirement $D_{\max}$:

$$(\mathrm{P2}) \quad \min_{\epsilon \in \mathcal{E}} \ \epsilon r_0 \quad \text{subject to} \quad J(\epsilon) \leq D_{\max}. \tag{16}$$

Figure 2 evaluates the mechanism and AoII mapping. Problems (P1) and (P2) specify alternative requirements. If conditional outages decrease with ϵ, the largest feasible budget solves (P1); no interior optimum is required. Otherwise, candidates are compared under identical link/source conditions, and the selected budget is independently confirmed.

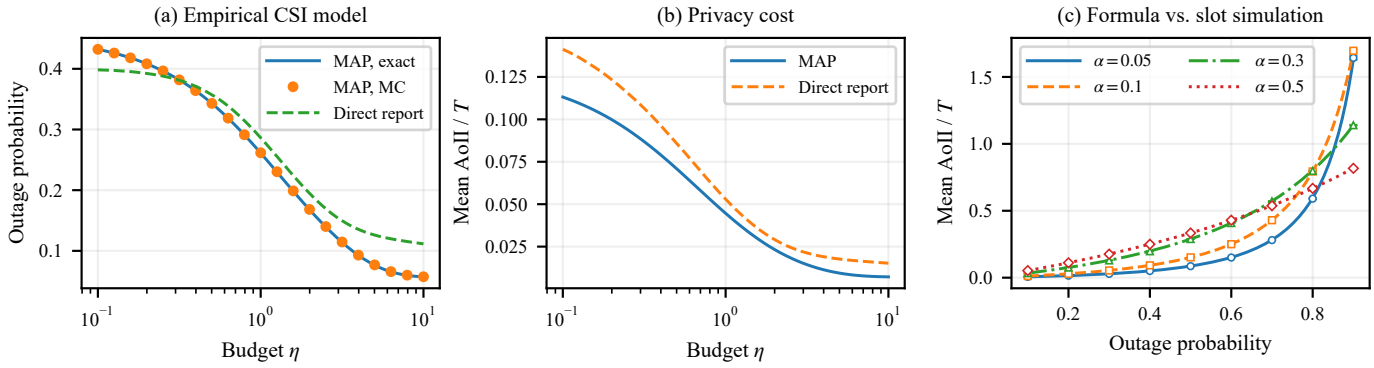


Fig. 2. Empirical-channel privacy cost and analytical validation. Solid MAP curves in (a) use analytical integration; markers and dashed direct-report curves use Monte Carlo (MC). In (c), lines are analytical and markers are direct task-state simulations with nominal pointwise 95% batch-mean intervals for 36 controlled settings.

Algorithm 1 LAPIS: Fixed Privacy-Budget Selection

Require: Public scenarios, prior/weights, grid $\mathcal{E}$, mode and constraint; $N_s, N_c, \delta_s, \delta_c$.
Ensure: Fixed $\epsilon^\star$, confirmation interval/status, and (P1) gap G; or no candidate.
Set $\mathcal{C} = \mathcal{E}$; for (P1), retain $\epsilon r_0 \le \eta_{\max}$.
If $\mathcal{C} = \varnothing$, **return** grid-infeasible.
Draw i.i.d. report-noise seeds and true-channel samples; reuse across budgets.
for each $\epsilon \in \mathcal{C}$ **do**
  Generate reports; apply MAP and the frozen beam map.
  Evaluate true-channel outages via (9); estimate $\widehat{q}_{\epsilon j}$ for each scenario by (17).
  Compute $[L_\epsilon, U_\epsilon]$ via (18)–(19).
end for
if mode is (P1) **then**
  Select $\epsilon^\star$ and compute G by (20); ties favor smaller budgets.
else
  Set $\mathcal{F}_D = \{\epsilon \in \mathcal{C} : U_\epsilon \le D_{\max}\}$.
  If $\mathcal{F}_D = \varnothing$, **return** infeasible when $\min L_\epsilon > D_{\max}$, otherwise unconfirmed; else set $\epsilon^\star = \min \mathcal{F}_D$.
end if
Freeze $\epsilon^\star$; evaluate with N_c new samples per scenario and radius d_c to obtain $[L_c, U_c]$.
return budget, interval, (P1) privacy status and G, or (P2) confirmation iff $U_c \le D_{\max}$; do not retune.

IV. LAPIS: Location-Aware Privacy–Information Selection

Algorithm 1 evaluates a finite grid and selects one fixed reporting budget. The grid, scenarios, prior, beam map, sample counts, and confidence levels are fixed before evaluation; no monotonic outage or convex objective is assumed.

A. Paired Outage Evaluation

For each scenario j and sample ℓ, draw $R_{j\ell} \sim \mathrm{Gamma}(3,1)$, a uniform unit-sphere direction $\mathbf{U}_{j\ell}$, and independent true-channel samples from (5) or the frozen empirical CSI bank. Reuse these draws across budgets, setting

$$\mathbf{z}_{j\ell}(\epsilon) = \mathbf{x}_j + \frac{R_{j\ell}}{\epsilon}\mathbf{U}_{j\ell},$$
$$\widehat{q}_{\epsilon j} = \frac{1}{N_s}\sum_{\ell=1}^{N_s}(1 - Y_{j\ell}(\epsilon)). \tag{17}$$

Each $Y_{j\ell}(\epsilon)$ is computed through (3)–(9) on the same true channel realization. This coupling preserves the marginal distributions and independence across ℓ. Let $F_\alpha(q)$ denote the right-hand side of (13). The plug-in estimate is $\widehat{J}(\epsilon) = \sum_j \omega_j F_\alpha(\widehat{q}_{\epsilon j})$.

To account for finite-sample uncertainty, Hoeffding's inequality [15] and a union bound over $N_{\mathrm{x}}|\mathcal{E}|$ pairs give the simultaneous radius

$$d_s = \sqrt{\frac{\log(2N_{\mathrm{x}}|\mathcal{E}|/\delta_s)}{2N_s}}, \tag{18}$$
$$q^-_{\epsilon j} = [\widehat{q}_{\epsilon j} - d_s]_+, \quad q^+_{\epsilon j} = \min\{1, \widehat{q}_{\epsilon j} + d_s\},$$

where $[a]_+ = \max\{0, a\}$ and $\delta_s \in (0,1)$. Since F_α is increasing, with probability at least $1 - \delta_s$, every candidate satisfies

$$L_\epsilon = \sum_j \omega_j F_\alpha(q^-_{\epsilon j}), \quad U_\epsilon = \sum_j \omega_j F_\alpha(q^+_{\epsilon j}), \tag{19}$$
$$L_\epsilon \le J(\epsilon) \le U_\epsilon.$$

These bounds cover Monte Carlo error under the assumed model, not map or channel-model mismatch.

B. Selection and Independent Confirmation

For (P1), form $\mathcal{F}_\eta = \{\epsilon \in \mathcal{E} : \epsilon r_0 \le \eta_{\max}\}$ and choose

$$\epsilon^\star \in \arg\min_{\epsilon \in \mathcal{F}_\eta} U_\epsilon,$$
$$G = U_{\epsilon^\star} - \min_{\epsilon \in \mathcal{F}_\eta} L_\epsilon. \tag{20}$$

On the simultaneous-coverage event, $J(\epsilon^\star) - \min_{\epsilon \in \mathcal{F}_\eta} J(\epsilon) \le G$; G bounds unresolved grid suboptimality.

For (P2), choose the smallest ϵ satisfying $U_\epsilon \le D_{\max}$. If none qualifies, $\min_\epsilon L_\epsilon > D_{\max}$ indicates grid infeasibility on the coverage event; otherwise feasibility is unconfirmed. An interval straddling the threshold leaves grid optimality unresolved.

Freeze the budget and use N_c independent samples per scenario, replacing d_s in (18)–(19) by $d_c = \sqrt{\log(2N_{\mathrm{x}}/\delta_c)/(2N_c)}$. For (P2), confirm feasibility only if the new upper endpoint is at most $D_{\max}$; otherwise report

TABLE I
SIMULATION AND EMPIRICAL-CHANNEL SETTINGS

Parameter	Value
Privacy distance / budget grid	$r_0 = 10$ m; 61 log-spaced $\eta \in [0.1, 10]$
Default task / threshold	$\alpha = 0.1$; $\gamma_{\mathrm{th}} = 1$; AoII in units of T
Empirical receiver / map cells	64 channels at 2.610 GHz; 10-m cells; 2-m guard
Empirical scenarios / weights	Three cell centers; uniform prior and weights
Configured empirical SNR scale	10 dB at median training channel energy
Mechanism Monte Carlo samples	3×10^5 per scenario and budget
Selection samples (main / sweep)	2×10^4 / 13 sizes, 200–20000
Confirmation samples	$N_c = 10^5$
Confidence / macro repetitions	$\delta_s = \delta_c = 0.025$; 1,000 repetitions
Geometric array control	$M = 1, 16, 64, 256$; $d_x = d_y = \lambda_c/2$
Geometric channel / scenarios	$\kappa = 10$, $\nu = 2$; 75 hovering locations
Geometric report samples	3×10^4 per location and budget

unconfirmed without retuning. The union-bound coverage for selection and confirmation is at least $1 - \delta_s - \delta_c$. For the geometry-based beam rule and brute-force MAP, selection costs $O(|\mathcal{E}|N_{\mathrm{x}}N_s(N_{\mathrm{x}} + M))$, followed by one-candidate confirmation. Figures 3 and 4 examine array sensitivity and selection reliability.

V. NUMERICAL RESULTS

A. Experimental Setup

Table I gives the settings. One 50-s recording from the MaMIMO-UAV dataset [16] yields 500 location–CSI pairs at 100-ms intervals on one subcarrier. Coverage is 34×43 m horizontally, at takeoff-relative heights 25–27.5 m. A 70/30 split of 10-m cells (seed 20261027), with a 2-m guard, retains 287 training and 102 test snapshots and discards 111. Seven covariance eigenvectors form the frozen nearest-cell beam map. Empirical SNR is $10|\mathbf{w}^{\mathsf{H}}\mathbf{h}|^2 / \operatorname{median}_{i\in\mathcal{T}} \|\mathbf{h}_i\|_2^2$, where $\mathcal{T}$ contains training snapshots. This configured SNR scale is uncalibrated. Ages and intervals are normalized by T, unrelated to measurement spacing.

Three occupied held-out cell centers define the public scenarios before reporting, with 7, 17, and 78 CSI snapshots. Snapshots are independently resampled within cells; scenario weights are equal. The privacy guarantee applies to representative centers, not original Global Positioning System (GPS) coordinates. Collinearity and equal priors permit exact MAP integration. This data-derived grid does not test general three-dimensional map performance. Bounds cover simulation randomness, not measurement representativeness, GPS error, or map mismatch.

B. Privacy Mechanism and AoII Validation

Figure 2 traces report privacy through the frozen beam map. Increasing η from 0.1 to 10 reduces mean outage from 0.4321 to 0.0571 and normalized AoII from 0.1131 to 0.0072. At $\eta = 1$, MAP achieves normalized AoII 0.0446 versus 0.0530 for direct-report map lookup, an estimated 15.8% reduction. Simultaneous 95% Monte Carlo bounds for the direct baseline give a relative-reduction interval of [13.9%, 17.6%]. MAP need not minimize outage: at $\eta = 0.1$, direct lookup has lower mean outage but higher mean AoII. This reflects heterogeneous outages and nonlinear scenario-wise AoII. For 36 controlled source–link settings, 5×10^6 simulated slots per setting give maximum relative deviation 0.78%; nine additional empirical-marginal settings give 1.33%. These tests assess the i.i.d. mapping, not measured-trajectory AoII.

C. Array Size and Directional Sensitivity

Figure 3 uses (5) to isolate aperture effects without assuming the measured array's physical ordering. Candidate coordinates are $x, y \in \{-100, -50, 0, 50, 100\}$ m and $z \in \{80, 120, 160\}$ m; conditional Rician outage is integrated analytically. At fixed power and $\eta = 0.1$, AoII rises from 0.821 for $M = 16$ to 2.378 for $M = 256$, despite the larger aligned gain. At equal reference aligned-SNR scale, the corresponding values are 0.326 and 3.152. Larger arrays therefore do not guarantee better privacy-constrained performance. The $M = 1$ curve is budget-independent, as expected. An additional analytical Rayleigh control is direction-independent (not plotted).

D. Budget Selection and Reliability

For (P1), LAPIS selects the largest feasible grid point for all four tested caps $\eta_{\max} \in \{0.3, 1, 3, 10\}$, matching the exact grid reference. These are boundary optima. For (P2) with $D_{\max}/T = 0.05$, it selects $\eta = 1$ and independently confirms normalized AoII in $[0.0431, 0.0459]$. At the tighter requirement 0.03, it selects $\eta = 1.848$, whereas the smallest feasible budget under the exact conditional reference is 1.585: finite-sample certification costs additional disclosure.

Figure 4 quantifies this tradeoff for $D_{\max}/T = 0.05$. With $N_s = 200$, point-estimate selection violates the exact empirical-model AoII requirement in 657 of 1,000 runs (65.7%), versus zero observed violations for LAPIS. The two-sided 95% binomial upper endpoint for LAPIS is 0.368%; zero observed violations do not imply zero risk. LAPIS selects mean budget 2.76 versus 0.85 for the comparator at this sample size; both move toward the conditional-model feasibility boundary as samples increase. All LAPIS candidates in this reliability experiment pass independent confirmation. Independent Bernoulli counts simulate reliability and confirmation; the selector never receives exact probabilities. At $\eta = 0.858$, AoII is 0.05031, just above the threshold, explaining persistent finite-sample misclassification on the denser grid. Broader measurements are needed for generalization.

VI. CONCLUSION

Location privacy can degrade receive-beam decisions and increase persistent task-information errors. The empirical and geometric experiments support this mechanism while showing that array size alone does not remove its cost. LAPIS translates privacy or AoII requirements into fixed reporting budgets with

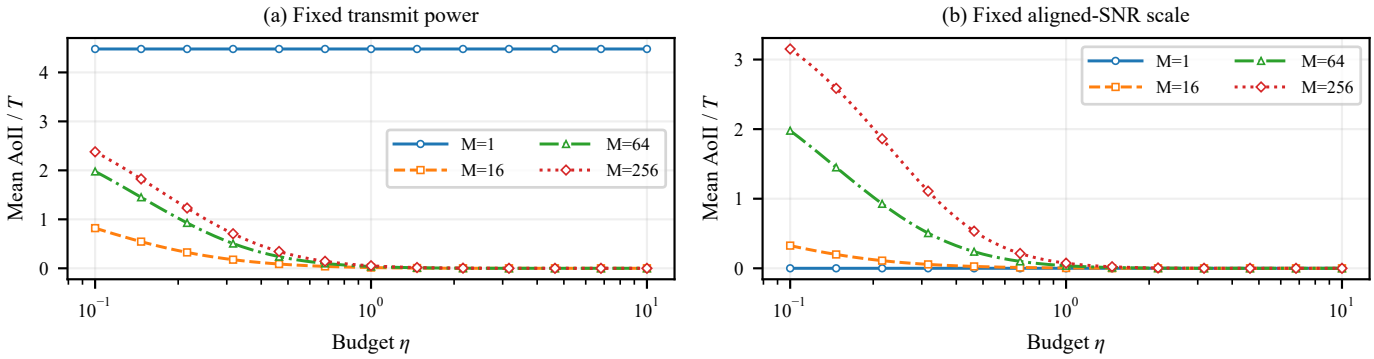


Fig. 3. Geometric Rician array controls. Fixed power uses reference single-element SNR -5 dB at 150 m. The second control fixes reference LOS aligned-SNR scale at 13.06 dB, adjusting power as $1/M$.

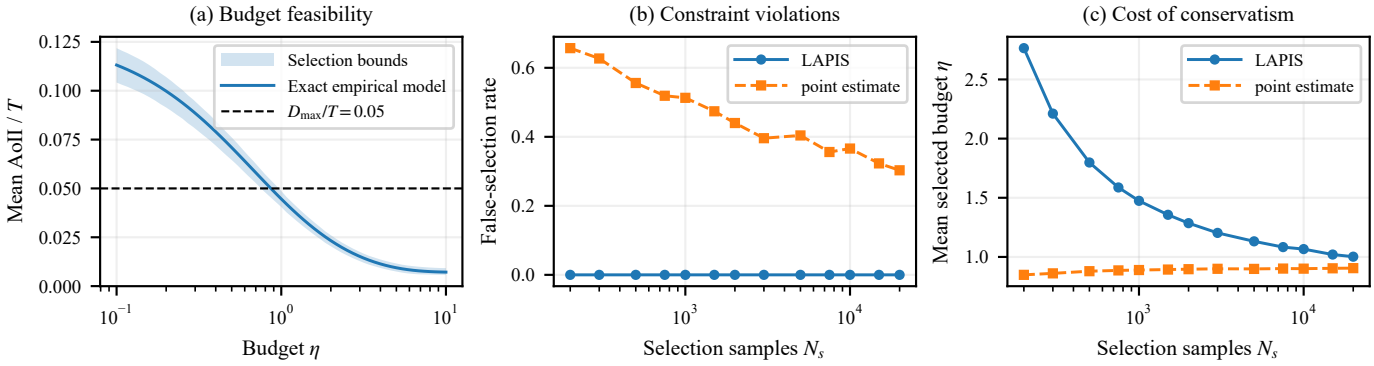


Fig. 4. Conditional empirical-model selection. Reliability uses equivalent Bernoulli links, independent budget streams, and 1,000 repetitions at each of 13 sample sizes. Both methods use identical selection samples and grids. Panel (b) reports selections that violate the exact-model constraint; panel (c) shows their disclosure cost.

explicit simulation uncertainty. Reliable selection in this conditional model requires additional disclosure at small sample sizes.

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